Status	Finished
Started	Thursday, 8 October 2026, 8:08 PM
Completed	Thursday, 8 October 2026, 8:14 PM
Duration	6 mins 7 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By dividing the list length by two ✓
c. By starting from the first element
d. By using a hash function

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Binary search
c. Linear search ✓
d. Hashing

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search stops immediately ✓
c. The search backtracks to find duplicate elements
d. The search starts over from the beginning

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Merge search
c. Quick search
d. Bubble search

Question **6**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(log n)
c. O(n log n)
d. O(1) ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **9**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The search is considered unsuccessful ✓
c. The last element is returned
d. An error is raised

Question **10**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓
b. Deleting elements from a list
c. Inserting elements into a list
d. Sorting elements in a list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. The list must be sorted
c. There should be direct access to the middle element in any sublist
d. Number values should only be present

Number values should only be present

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The search continues in the left or right sublist ✓
d. The list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 90 and 99 ✓
c. 89 and 94
d. 94 and 99

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity
b. Insertion
c. Selection
d. Bubble ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(1)
- c. O(log n)
- d. O(n log n)